

Yuchen Liu

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AI systems engineer building RL training infrastructure, data/search pipelines, and research-grade open-source tools.

Education

University of Pennsylvania, M.S. Computer Science, GPA 3.80/4.00 (May 2025)
Pennsylvania State University, B.S. Applied Economics, GPA 3.86/4.00 (May 2023)

Experience

Software Engineer, Search Infrastructure | DoorDash | Aug 2025 - Present | Seattle, WA

- Built **Notus**, a next-generation search ingestion platform with Iceberg, Kafka, and Spark, decoupling document ingestion from serving and enabling configuration-driven indexing into multiple backends.
- Built **Vela**, Argo's hybrid retrieval platform, exposing Lucene-native hybrid search and federated vector retrieval with backends such as Milvus behind one API.
- Migrated 30+ production stacks to modular Jsonnet, eliminating roughly 56K lines of duplicate configuration and reducing config verification time by 90%.
- Resolved indexer OOM and disk-capacity incidents with 500GB EBS storage, pre-deployment validators, and Kyverno guardrails; led RCA work for revenue-impacting reliability failures.

Software Engineering Intern | Penn Medicine TissueLab | Oct 2024 - May 2025 | Philadelphia, PA

- Developed a Python/FastAPI microservice platform to orchestrate PyTorch models for medical image analysis, enabling natural-language-driven segmentation and classification workflows.
- Implemented an event-driven DAG workflow engine for async task execution, processing 10K+ pathology images daily and streaming 12M+ imaging events with sub-10ms latency.

Software Engineering Intern | Information Technology of CAS | Apr 2024 - Aug 2024 | Remote

- Designed Kafka and Redis-backed event streaming services that reduced p95 API latency from 2s to 200ms while processing millions of IoT events.
- Contributed to distributed Spring Boot monitoring services, delivered 35+ REST APIs, and maintained 99.7% uptime in Unix-based production environments.

Open Source

nanoRL | github.com/RiddleHe/nanochat

- Led an open-source RL training framework inside nanochat for clean objective definitions, RL algorithm research, reproducible ablations, and production-grade vLLM inference serving.

nanochat | github.com/RiddleHe/nanochat

- Built a hackable pretraining and chat stack supporting architecture definition and FLOP-controlled model ablations.

llm-interp | github.com/RiddleHe/llm-interp

- Authored reproducible interpretability scripts for model circuit research, including attention-sink analysis and reproductions of LLM decode indeterminism findings.

Research

Do Value Vectors in Deep Layers Need Context from the Residual Stream? | EMNLP 2026, under review

Co-first author. Proposed context-free value vector tables for deep layers, eliminating the V cache and improving validation loss and average scores across 21 benchmarks. [arXiv:2606.02780](https://arxiv.org/abs/2606.02780)

Vision Language Models Cannot Plan, but Can They Formalize? | ECCV 2026, under review

Studied whether VLMs can formalize visual planning tasks into solver-executable PDDL problem files. [arXiv:2509.21576](https://arxiv.org/abs/2509.21576)

A co-evolving agentic AI system for medical imaging analysis | Nature, under review

Built agentic medical-imaging workflows for tool selection, planning, knowledge updates, and human-in-the-loop co-evolution. [arXiv:2509.20279](https://arxiv.org/abs/2509.20279)

Skills

Languages: Python, Kotlin, Java, C++, JavaScript **ML/Infra:** PyTorch, vLLM, Spark, Kafka, Iceberg, Delta Lake, FastAPI, Redis, Docker, Kubernetes, AWS, Spring Boot